25900554 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCPHY200 - PRINCIPLES OF MECHANICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Write the equation of motion for a body falling through air under the action of the force of gravity. [K] / [CO1]
2. How is gravitational mass different from inertial mass? [K] / [CO1]
3. Consider an object of mass $m = 1\ kg$ that is initially at rest and at the origin. Determine the velocity of the object when it reaches $x = 3\ m$ if it is acted upon by a force $F = 4x$. [A] / [CO1]
4. How does the gravitational force between two bodies change if the distance between them is doubled? [U] / [CO2]
5. If masses are present in a region, the gravitational potential cannot be zero, whereas the gravitational field can be zero.Comment on this statement. [U] / [CO2]
6. What are the differences between gravitational force and gravitational field? [U] / [CO2]

7. Why is the motion of a pendulum becomes simple harmonic motion for small angles only? [U] / [CO3]

8. Write the general solution for the overdamped case of a damped oscillator. [K] / [CO4]

9. A stone tied to a string is rotated in a horizontal circle of radius $0.75\ m$ with angular velocity $\vec{\omega} = 5\ rad/sec$. Calculate the centrifugal force on the stone of mass $0.25\ kg$. [A] / [CO6]

10. State and explain conservation of angular momentum with an example. [U] / [CO6]

11. Explain the significance of conservation of angular momentum in the case of Earth-Moon system. [U] / [CO6]

12. Is coriolis force affected the weather on the earth? Justify [U] / [CO6]

13. What is the relation between a central force and the corresponding potential energy? [U] / [CO7]

14. What is effective potential energy in central force motion? Illustrate it with energy diagram for an attractive inverse square law field. [U] / [CO7]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15. A particle of mass $2\ kg$ is acted upon by three forces (in Newtons) $F_1 = 2\ \hat{i}$, $F_2 = 6\ \hat{i} + 4\ \hat{j}$ and $F_3 = 10\ \hat{k}$. Find the acceleration of the particle. [A] / [CO1]

16. a) Draw the gravitational field lines due to an isolated point mass.
b) Sketch the corresponding equipotential surfaces.
c) Explain the nature of these field lines and equipotential surfaces. [U] / [CO2]

17. A particle performs simple harmonic motion with amplitude $5\ cm$ and time period $2\ s$. At what displacement is its speed half of the maximum speed? [U] / [CO3]

18. What is a damped harmonic oscillator? Write the equation for damping term. Obtain the differential equation of damped harmonic oscillator. [U] / [CO4]

19. Obtain the particular solution of a forced damped harmonic oscillaor. [U] / [CO5]

20. Derive an expression for angular momentum of rigid body about a fixed axis. [U] / [CO6]

21. Derive the relation $L = L_c + L'$, where $L_c$is angular momentum of the center of mass and L' is angular momentum relative to the center of mass. [U] / [CO6]

22. State Kepler's laws of planetary motion. [K] / [CO7]

[5x6 = 30]